Midterm Review

ECON 97: Economic Toolkit

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iClicker course link

https://join.iclicker.com/RDNG

Review of Concepts - Week 1

- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
- Compliment: $P(A') = 1 - P(A)$
- Mutually exclusive: $P(A \cap B) = 0$
- Exhaustive: $P(A \cup B) = 1$
- $P(A \cup B)' = P(A' \cap B')$ and $P(A \cap B)' = P(A' \cup B')$
- $P(A \cap B') = P(A) - P(A \cap B)$
- Subset: if $A \subset B$ then $P(A \cap B) = P(A)$

Review of Concepts - Week 2

- Replacement, order matters: n^k
- No replacement
  - Order matters: ${}_nP_k = \frac{n!}{(n-k)!}$
  - Order doesn't matter: ${}_nC_k = \frac{n!}{(n-k)!k!}$
- ${}_nC_k =_n C_{(n-k)}$
- ${}_nC_n = 1$
- ${}_nC_1 = n$
- ${}_nC_0 = 1$

Review of Concepts - Week 3

- $P(A|B) = \frac{P(A \cap B)}{P(B)}$
- $P(B|A) = \frac{P(A \cap B)}{P(A)}$
- If A, B independent ($A \perp B$):
  - $P(A \cap B) = P(A)P(B)$
  - $P(A|B) = P(A)$ and $P(B|A) = P(B)$
  - $A' \perp B'$, $A' \perp B$, and $A \perp B'$
  - $A \perp B \perp C \implies P(A \cap B \cap C) = P(A)P(B)P(C)$

Review of Concepts - Week 4

- **PMF**: $P(X = x) = f(x)$
  - E.g., $f(3) = P(X = 3)$
  - $\sum_{S_X} f(x) = 1$
  - Support of X (S_X): all possible values of X
- **CDF**: $P(X \leq x) = F(x)$
  - E.g., $F(2) = P(X \leq 2) = f(0) + f(1) + f(2)$
- **Mean**: $\mu = \sum_{S_X} x f(x)$
- **Variance**: $\sigma^2 = \sum_{S_X} (x - \mu)^2 f(x) = \sum_{S_X} x^2 f(x) - \mu^2$

Review of Concepts - Week 5

- Expectations
  - $\mu = E[X] = \sum_{S_X} xf(x)$
  - $E[g(X)] = \sum_{S_X} g(x)f(x)$
    - E.g., $E[X^2] = \sum_{S_X} x^2 f(x)$
  - $\sigma^2 = E[(X-\mu)^2] = E[X^2] - (E[X])^2$
- Properties
  - $E[c] = c$
  - $E[cX] = cE[X]$
  - $E[X \pm Y] = E[X] \pm E[Y]$
  - $E[aX \pm bY] = aE[X] \pm bE[Y]$